1 Executive Summary

This proposal outlines the University of Hong Kong (HKU) Design, Build and Fly team's plan for designing, building, and flying the "Griffin" aircraft to compete in the 2025-2026 AIAA competition. The competition consists of four missions focused on passenger and cargo carriage and a mechanism to stow, tow, deploy, and release a banner remotely. The objective is to design, build, and fly a remote-controlled fixed-wing aircraft to simulate payload carriage and banner towing. Weekly meetings and a Gantt Chart will assist in keeping the team organized and on schedule. Performance in the competition is based on payload, lap time and the successful remote deployment and release of a banner.

To maximize the overall score, the team conducted a score analysis, prioritizing the highest possible payload weight (W_p) and the longest banner length (B_L), while factoring in optimal lap count and time. The analysis led to the following mission targets: 15.43-lbs W_p and 36-s lap time for Mission 2 (M2); 80-inch B_L and six laps completed for Mission 3 (M3). Ground mission (GM) time is expected to be kept within 300-s by maximizing the W_p. The design was further refined through a series of trade studies, resulting in a high-wing monoplane configuration with a tractor motor, conventional tail, and tricycle landing gear. The aircraft features a rectangular wing with a 5-ft wingspan, a Wortmann-FX airfoil with a 0.91-ft chord at the inboard region where flaps are installed, and Clark-Y aileron at the outboard region where ailerons are installed. The fuselage measures 5-ft in length, 0.67-ft in width, and 0.58-ft in height. The propulsion system was optimized to maximize thrust given the estimated payload weight, with a battery capacity of 93.24 Wh. A comprehensive manufacturing plan was developed, detailing next steps in design, CAD modelling, electronics selection and both simulation and physical testing. The testing outline includes a range of tests intended to validate the design and simulation models, with specific deadlines to monitor progress. The highest score the team is estimated to obtain is 5.61 out of 6.

2 Management Summary

2.1 Team Organization

The HKUDBF 2025-26 team consists of twelve undergraduate students, supported by a Faculty Advisor who provides guidance during regular design reviews. The team leader handles both logistical and technical responsibilities, organises meetings, and arranges travel from Hong Kong to the US. The team leader also manages the meeting documents with the administrative officer, the budget with the financial officer, and the progress timeline with the project manager to ensure that team deadlines are met. In addition to this, the chief engineer oversees the integration of the design process and testing of aircraft components, with coordination and adjustments discussed with all the division-leads. The team is divided into 6 divisions as shown in Figure 1. Each division has a designated leader who is responsible for distributing tasks, organising sub-team meetings, and ensuring progress is well-documented. The specific responsibility of each division is detailed in Table 1.

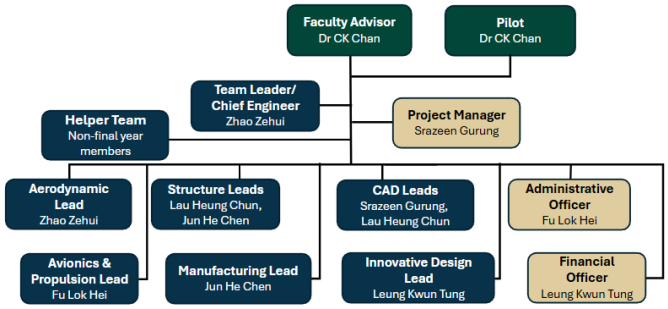


Figure 1 Team's Organisation Chart

Division	Responsibility	Skills needed
Aerodynamics	Aerodynamics analysis for sizing and desired performance	Fundamental Aerodynamics, derive parameters, CFD simulation
Propulsion & Avionics	Motor and propeller selection, electronics selection for stability and control	Analysis of required and available power; electronics, stability and control theory
Structure	Design airframe and detail structure and conduct testing to ensure structural stability	Structural analysis and FEA knowledge
CAD	Model design and modification	CAD and FEA knowledge
Manufacturing	Manufacture the aircraft	Machine tools and material property analysis.
Innovative Development	Testing on various innovative materials	Knowledge on 3D printing techniques and 3D print materials

Table 1 Sub-team Responsibility and skills

2.2 Schedule

The team's proposed schedule is depicted in the Gantt chart in Figure 2 and is reviewed during weekly meetings to ensure the timely completion of objectives and milestones. To facilitate effective communication between divisions, meetings are held according to teammates' availability to allow for effective, cross-functional collaboration across the teams.

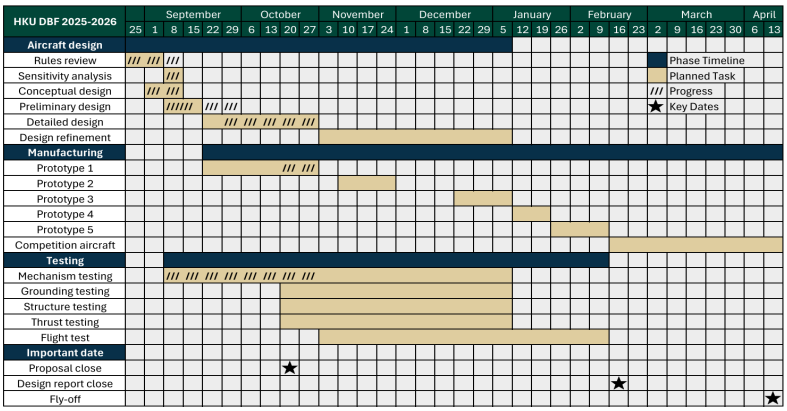


Figure 2 2025-2026 Project Gantt Chart

Category	Item	Cost Toal
Manufacturing	3D Printing Materials	$400
	Tools	$1000
Propulsion, Control, & Electronics	Servo Motor, Receiver, Transmitters, Batteries	$1431
	Motors, Propellers	$880
	Landing Gear	$200
	Misc. Electronic Components	$100
Structures	Prototyping Materials (Wooden Board/Carbon Fiber etc)	$650
	Glues & Tapes	$200
	Part Commissions	$150
Mission Systems	Payload & Banner system	$330
	Banner Parts	$260
Flight Test	Miscellaneous	$1000
Travel	Administrative	$350
	Dining	$1100
	Flight & Luggage (with shipping)	$11800
	Accommodation	$4000
	Car rent, vehicle fuel & insurance	$1500
	TOTAL COST (USD)	**$25001**

Table 2 Budget Plan

2.3 Budget

HKUDBF's 2025 anticipatory budget is shown in Table 2. The budget encompasses costs associated with manufacturing, materials, electronics, administration (including membership fees) and travel (including shipping the competition plane via air). The gross proceeds total $25001, funded by the Innovation Wing of the Faculty of Engineering ($20231) and the Department of Mechanical Engineering ($4770). Expenses are reduced by utilizing existing resources on campus, including manufacturing labs, tools and machines. The team is currently operating with a 20% budget margin.

3 Conceptual Design Approach

3.1 Analysis of Mission Requirements

The competition consists of three flight missions and one ground mission, themed around payload carriage and a "banner-towing business". With the sensitivity analysis result for M2, the aircraft design should prioritize optimal payload weight and battery capacity while minimising flight time. For M3, the focus shifts towards finding a compromise between the banner size and drag induced by it, while minimizing wingspan and lap time to achieve the largest possible aircraft speed and banner length.

Mission	Scoring	Description	Subsystem Requirements
M1	$M1 = 1.0$	· Ready aircraft within 5 minutes. · Complete three laps within 5 minutes	· The aircraft must be durable and reliable to achieve stable flight under varying conditions.
M2	$1 + \frac{(Net_{Income})_{Team}}{(Net_{Income})_{Max}}$	· Attach payload and ready aircraft within 5 minutes. · Carry passengers and cargo onboard.	· Aircraft must optimise payload weight carried and minimise lap time. · Aircraft must be strong enough to carry maximum payload weight.
M3	$2 + \frac{\left[\frac{Laps\ Flown \times B_L}{RAC}\right]_{Team}}{\left[\frac{Laps\ Flown \times B_L}{RAC}\right]_{Max}}$	· Assemble and checkout in 5 mins. · Release banner by remote command after crossing the finish line on the last lap. · Fly as fast as possible	· Banner mechanism must work properly to stow, deploy, and release the banner. · Aircraft must take account of moment induced by banner.
GM	$\frac{(Mission\ time)_{min}}{(Mission\ time)_{Team}}$	· Ready control surfaces · Load payload as fast as possible · Perform banner towing mechanism successfully	· Beware of the mechanism control and time for the payload loading and unloading process
Total score: $(\boldsymbol{M1 + M2 + M3 + GM}) \times (\boldsymbol{Report} \times 85\% + \boldsymbol{Proposal} \times 15\%) + \boldsymbol{Participation\ Score}$			

Table 3 2026 Mission Summary

3.2 Trade Studies

To optimize the baseline design of the aircraft, trade studies were performed and encapsulated in Table 4 with critical performance matrices. Each option was rated on a scale of -1 to 1, with 1 being the best possible. A rectangular wing eases manufacturability and best complements subsonic aircraft design. A hybrid airfoil merging Clark-Y and WFX airfoils in the wing was adopted for benefits in both stall stability and enhanced lift force. A trapezoidal empennage is a tried-and-tested stabilizer design that meets the overall demands of all missions, integrating performance factors and reliability in a 5-minute flight window. A tractor design is selected for its ease of manufacturing, favourable balance between stability and aerodynamic drag, and sufficient ground clearance. A servo lock mechanism is mounted beneath the fuselage, with the banner thread secured to the

tail arm to reduce both deployment time and overall weight. To enhance loading efficiency and maximise space utilisation, a double-deck payload compartment system was implemented, ensuring effective transport of the designated payload.

Main Wing Planform and Airfoil	Weighting	Swept Back	Rectangular	Tapered	Clark-Y	Wortmann FX 63-137 (WFX)	Hybrid of 2
Manufacturability	20%	-1	1	-1	1	-1	0.5
Weight	20%	1	0.5	0.75	-0.5	0.75	0.5
Flight stability	25%	1	0.5	1	0.5	1	0.75
Lift	15%	0.5	0.25	0.25	0.75	1	0.75
Drag	10%	1	0	0.5	0.75	1	0.75
Structural Complexity	10%	0	1	0	1	-1	0.75
Total Score	100%	0.53	0.54	0.43	0.44	0.44	0.65

Empennage	Weighting	Rectangular	Tapered	Trapezoidal
Manufacturability	10%	1	0.75	0.75
Weight	20%	1	0.75	0.75
Stability	30%	-0.5	0.75	0.75
Drag	25%	-1	0.5	0.75
Mission Sustainability	10%	0.5	0.75	1
Scoring (M3)	5%	0.25	0.75	1
Total Score	100%	0.62	0.69	0.78

Motor	Weighting	Tractor	Pusher	Wing Mounted	Top Mounted
Manufacturability	10%	0.7	0.2	0	0
Weight	15%	0	0	-1	-0.2
Stability	20%	0.6	-0.5	0.3	-0.5
Drag	20%	0	1	0.5	1
Ground Clearance	10%	0.8	−1	0	1
Propeller Efficiency	15%	0.8	0.6	0.5	1
Noise and vibration	10%	0.6	0.4	0.3	0.2
Total Score	100%	0.45	0.15	0.115	0.34

Banner mechanism	Weighting	Servo Lock	Spring Lock	Spindle Lock
Manufacturability	20%	1	0.5	0.5
Structural Complexity	25%	0.5	0	0
Deployment Speed	20%	1	0	-0.5
Deployment Success	35%	0.5	0.75	0.75
Total Score	100%	0.7	0.36	0.26

Payload System	Weighting	Double Deck	Single Deck
Manufacturability	20%	0.5	1
Structural Complexity	20%	0.5	1
Spatial Efficiency and Feasibility	35%	1	0.5
Loading Speed	25%	1	0
Total Score	100%	0.8	0.575

Table 4 Trade Studies

3.3 Sensitivity Analysis

The sensitivity analysis aims to evaluate the influence of key design and mission parameters on the performance of an aircraft in various mission scenarios. By identifying the parameters with the greatest impact on performance, the study helps guide design decisions and optimize aircraft capabilities for different operational conditions to maximize team's score, which is expected to be 5.61 out of 6. **The analysis was conducted on MATLAB by varying parameters such as the number of passengers, cargoes, and laps, as well as the battery capacity, banner length and rated airplane cost (RAC), evaluating their effect on different mission scores.** Sensitivity of each parameter was assessed using a graphical representation of parameter deviations, their corresponding impact on performance metrics, and the steepness of the trend. In Figure 3, by graphical interpretation, all parameters increase linearly with the Mission 2 score, except for battery capacity. The number of laps has the steepest gradient among all parameters, suggesting that this is the most sensitive factor for Mission 2's scoring. This is followed by the number of cargoes, number of passengers and battery capacity. However, the team needs to demonstrate the plane's capability to carry at least 3 passengers for each piece of cargo in the technical inspection even though the initial wish was to maximize the number of laps by carrying fewer passengers and cargo. To determine which parameter to maximise, the team should account for the effect of changing the number of passengers and cargos simultaneously rather than changing the two variables individually. By combining the two graphs, an increase of 50% in the number of cargoes and passengers yields a 6.99% higher score compared to a 50% increase in lap count. **Therefore, the team chose to maximize the number of payloads in the design.** In Figure 4, on one hand, both the banner length and number of laps have the same linear relationship with the M3

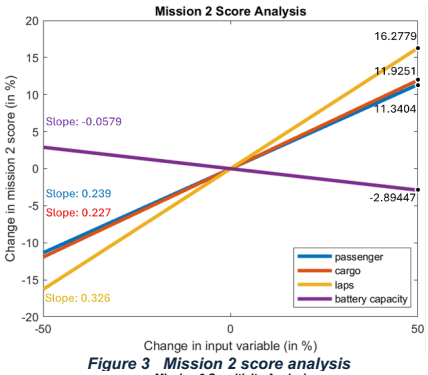


Figure 3 Mission 2 score analysis

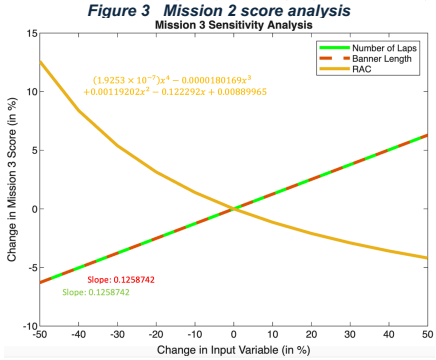


Figure 4 Sensitivity Analysis M3

score, so maximising either would be beneficial, although admittedly, increasing the banner length is more readily feasible. On the other hand, the RAC of the aircraft has a relationship with the M3 score that is best described with a quartic regression. This relationship states that with a lower RAC, the score shoots up considerably. **With that observation in mind, the team minimized the wingspan of the aircraft to lower the RAC while ensuring the aircraft generates enough lift to complete all missions.**

3.4 Design Optimization Approach

To determine the best aircraft design, factors such as payload weight, dimensions and locations of wing and tail, wing airfoil, banner drag and battery power were set as dimensional constraints for DBF's missions scoring formulae. The aircraft's centre of gravity (CG), performance and stability were calculated for every iteration on these constriants for the design model, and the resulting scores were compared to see which yielded the highest total mission score possible on MATLAB and EXCEL. The results are further corroborated empirically and physically by comprehensive ground and flight tests.

3.5 Preliminary Design

3.5.1 Griffin Design

Figure 5 displays the CAD model of Griffin. Given a competition design constraint permitting a maximum wingspan of 5ft, the team's design optimization approach adopts a 4.95-ft wingspan as a buffer against manufacturing errors. The rectangular wing has a chord of 0.9-ft with no taper to maximize the wing area and lifting force for M2, while easing manufacturability. An aspect ratio of 5.5 was chosen to optimize the lift over drag ratio for M3. Through the optimization approach, the hybrid wing with Clark-Y as the outboard region and the Wortman FX 63-137 as the inboard region was selected to promote sufficient stall stability and a high lift coefficient, which computes to 0.581 when the AoA is at 0 degrees. A conventional trapezoidal tail with a horizontal stabiliser of 2.92 ft in length (0.71 ft^2in area), and a vertical stabiliser of 2.50 in length (0.36 ft^2 in area), was selected for better stability and control of the aircraft. A truss-structured fuselage of 0.67-ft in width and 0.58-ft in height is sized to fit all payloads, the 6S 4200mAh (93.24-Wh) 55C battery, a 120A ESC, a receiver, a 490Kv motor, and a 16"x10" propeller amongst other components. **The empty weight of Griffin is estimated to be 7.72-lbs, with a maximum take-off weight of 15.8-lbs**. The aerodynamic centre is indicated by the yellow rhombus in Figure 5. The impact on stability by the payloads and banner are examined carefully in all theoretical calculation, ground testing, and flight-testing phases.

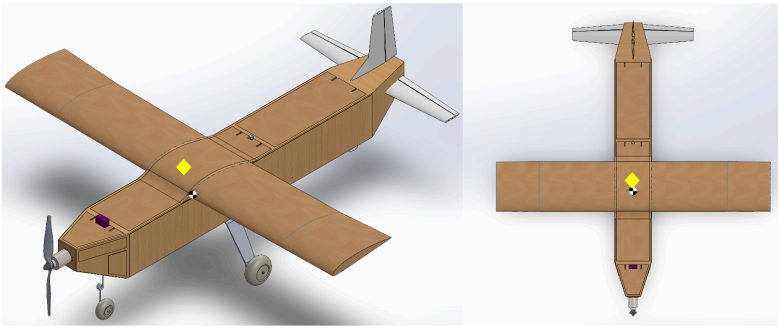


Figure 5 Aircraft "Griffin" Assembly CAD

3.5.2 Payload Constraint System Design

The aircraft must carry 3 passengers for every cargo declared in the technical mission. The payload compartment cannot be mechanically fastened by frictional force. The payload compartment design was driven by the M2 sensitivity analysis that identified payload amount as the primary scoring variable. The competition rules mandating at least three ducks per puck shifted the team's objective from maximizing payload to optimizing spatial efficiency. **Subsequent analysis determined the optimal payload to be 16 hockey pucks and 48 ducks**, which directly defined the core airframe configuration. **To accommodate this payload, a three-duck-per-row arrangement was selected after a conducted trade study.** Configurations with one or two ducks per row would require an excessively long fuselage, which increases susceptibility to bending and torsional stress. A four-ducks-per-row layout would shorten the fuselage and increase the plane's cross-sectional area, increasing drag and complicating weight distribution. To seat all 48 ducks, a dual-deck architecture was adopted.

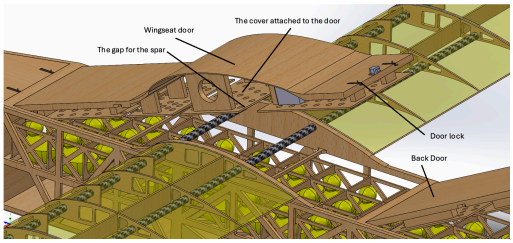


Figure 6 Passenger Compartment and Retainment System

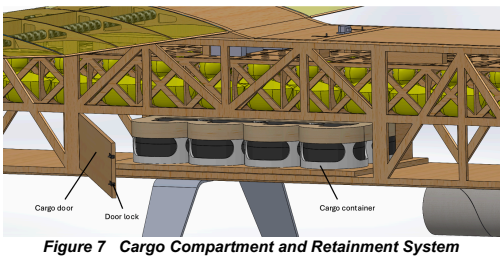


Figure 7 Cargo Compartment and Retainment System

This places the heavier ducks in the lower cargo bay, favourably positioning the CG, with the lighter ducks on the upper deck. In this way, the CG positioning for M2(with payload) and for M3 (without payload) would vary slightly but in an acceptable range. For the ground mission, the design emphasizes simplicity and reliability by integrating the passenger container into the airframe. **Ducks will be loaded through special cover lid-integrated doors on the fuselage roof and are secured automatically when the doors close.** The cover lid uses a sandwich structure with balsa wooden boards on top and foam beneath, providing secure constraint with allowance for the ducks' height while adding rigidity. Hockey pucks will be loaded into a specialized container with a wood or foam cover for strength, retainability, and weight minimization. **This hockey puck container is loaded through a side cargo door and is retained automatically by a simple built-in holder that locks it in place when the container slides in the slot.** These approaches limit the degree of freedom of the ducks and pucks while simultaneously minimizing weight and complexity. They also present a practical and robust solution for a rapid ground mission.

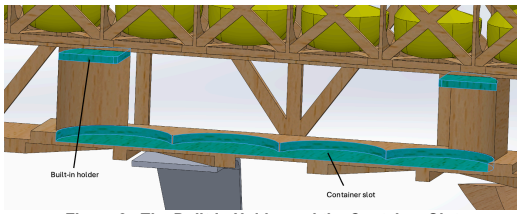


Figure 8 The Built-In Holder and the Container Slot

3.5.3 Banner Mechanism Preliminary Design

The banner mechanism must perform the stow, deploy, tow and release function successfully in M3. The ratio of the banner length to the banner width must be ≤ 5 with a minimum length of 10 inches and height of 2 inches, and the banner must have the team's university name or logo on both sides. To stow the banner externally to the airplane, the preliminary design is made as shown in Figure 9, and 10. A small square wooden stand with a hole is extruded on the bottom side of the fuselage. A pin will be inserted into the stand and fixed by tolerancing methods. A thread will be tied from the pin to the servo arm. This servo is placed securely in an opening on the bottom side of the fuselage, fixed by machine screws and wooden blocks. The servo is controlled by an additional transmitter. The threads will wrap the stowed banner tightly under the fuselage, and the pin will be released once the servo rotates from $0°$ to $110°$, deploying the banner. The towed banner is shown in Figure 12. A carbon-fibre rod with a one-sided weight will be attached on the left side of the banner, with the weight pointed downward to make sure the banner stays in the upright position. The Y-shaped thread of the banner is fastened to a circular pin which is locked by an L-shaped hook inside the fuselage as shown in Figure 13. Another servo placed inside the fuselage turns the hook from $90°$ to $110°$, removing the long circular pin which releases the banner from the aircraft completely. To reduce the drag induced by the banner and ensure successful performance, several methods and materials will be tested for the mission. The tentative banner design is a mesh banner which contains tiny holes, allowing air to pass through. The team aims to carry an 80-in by 16-in banner in M3 which maximizes the score the team can achieve. The team's university logo and names are printed on both sides of the banner with a weight of 0.77-lbs. The possible locations where the banner may be stowed with negligible impact on the aircraft's CG are shown by blue dots in Figure 12. The impact of the banner is further quantified by mathematical formulae, and all design choices will be tested and improved after critical data, such as banner drag and CG, are obtained from flight tests.

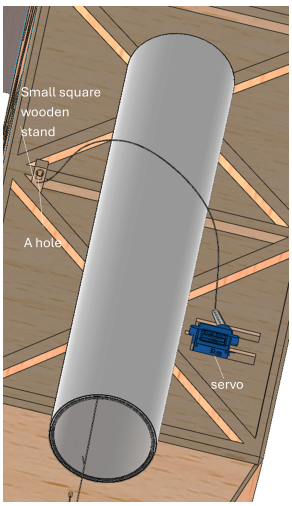
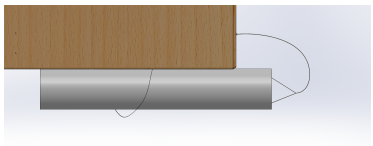


Figure 9 Banner attachment mechanism


Figure 10 Stowing stage

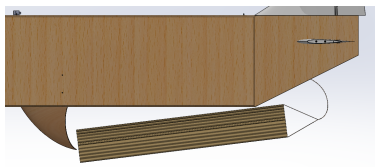

Figure 11 Deploying stage



Figure 12 Towing stage

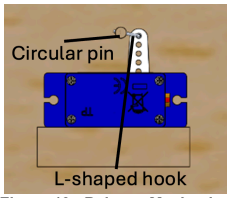
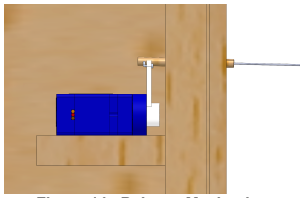


Figure 13 Release Mechanism Front View


Figure 14 Release Mechanism Side View

4 Manufacturing Plan

The manufacturing process is tightly integrated with design and simulation to ensure performance and reliability. Following mission analyses and CAD modelling, CFD and FEA simulations guide material selection and structural design as well as validate aerodynamic and structural performance. Manufacturing streams run in parallel Commercial components are sourced: laser cutting for airfoil ribs and bulkheads, 3D printing or laser-cut wooden truss modelling for payload bays and prototyping, and metal bending for landing gear and structural reinforcements. Commercial components are sourced early to streamline integration. Initial prototypes validate geometry, CG, and subsystem integration. Subsequent prototypes and test flights informs iterative improvements, with spare parts pre-manufactured to support rapid turnaround. This plan emphasizes modularity, precision, and readiness for competition conditions. A feedback loop enables rapid design modifications, ensuring readiness for competition.

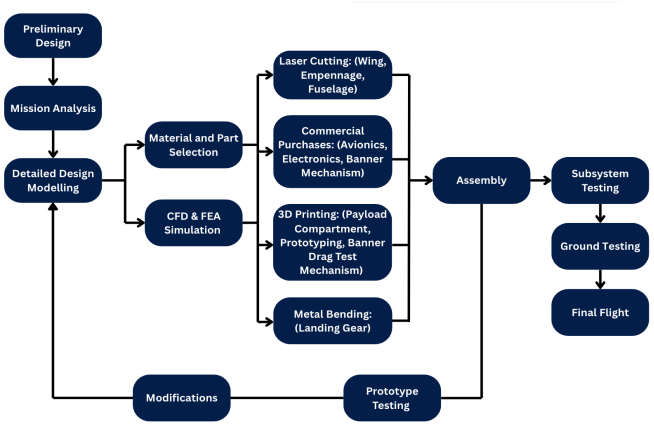


Figure 15 Manufacturing Plan

5 Testing Plan

The testing plan shown in Table 5 shows the tentative schedule for various design validation stages after the preliminary design to verify and finalise the design choices. The propulsion test evaluates the performance of the motor in an open area with static airflow using a propulsion testing stand rather than using a wind tunnel which limits the testing space. Ground, structural, payload and banner tests will provide critical data for verifying the structural rigidity, functionality, assembly time and induced drag of the aircraft before flight testing. To collect the flight data needed to quantify the aircraft's performance, a Pixelhawk flight controller and camera will be equipped to record required variables such as acceleration, velocity, stability, climb/turn rates, flight distance, etc. These electronic components will be removed before the Fly-off to satisfy competition regulations. A series of documentation and apparatus will be used to comprehensively record and report the results of all flight tests, including safety checklists, methodology reports, camera footage, and after-action reports. The overall performance will be assessed by all team members and pilots, followed by group reflections on areas for improvements.

Division	Test	Method	Objectives	Due
Propulsion Test				
Avionics	Static Thrust Test	A propeller test stand to assess thrust, battery current, power, and RPM across various propeller and motor configurations; obtain static thrust values.	Aerodynamic performance validation, sufficient thrust generation.	Nov, 25
Ground Test				
Structure, CAD	Restraint system test	Adding payloads to the restraint systems and record the highest holdable amount.	Structural integrity under high G-force.	Nov, 25
Aerodynamic, Avionics	Banner mechanism test	Perform the release mechanism of the banner using a transmitter, receiver, and servos.	Validation of the banner mechanism.	Nov, 25
Aerodynamic, Avionics	Banner drag test	Perform drag analysis of banner with various sizes and materials; test and conduct actual banner drag test on a prototype with flight controller. Possible attachment locations are shown as blue dots in Figure 12.	Measurement of drag to enhance score analysis accuracy.	Oct, 25
Structure, CAD	Wing Tip Test	Add payloads to the aircraft to its maximum weight and lift it vertically by the wingtips. Elevate the aircraft from the wingtips at the calculated CG point and verify that it remains level throughout the mission.	Validation on wing attachment and structural integrity and the appropriate CG position.	Nov, 25
Manufacturing, structure, avionics	Assembly Test	Simulate each ground mission and log the corresponding times required for completion. Repeat the process until the target times are achieved. Various payload loading and unloading mechanism methods will be tested.	Verification of the efficiency of the assembly method and minimization of ground mission time.	Jan, 26
Avionics	Electronics test	Connect all electronic components and validate their functionality, including independent servo calibration, propulsion, control surfaces, and failsafe systems. Perform range check.	Proper function of the electronic components and fulfilment of the mission requirements.	Oct, 25
Aerodynamics	Aerodynamic Analysis	Use XFLR5 and Ansys to assess the aircraft's stability characteristics and banner drag performance.	Validation of the theoretical values of the aircraft stability.	Nov, 25
Manufacturing, structure	Landing gear test	Perform a loading test and inspect the landing gear and associated connections for any damage.	Validation of landing gear configuration and strength for take-off and landing.	Nov, 25
Structure	Structural test	Use FEA methods to assess individual components and whole aircraft's structural rigidity.	Validation of structural rigidity before manufacturing.	Oct, 25
Flight Test				
All leads	Missions and flight test	Perform M1, M2, M3 and GM, check the installation of all parts and assembly.	Completion of all missions and achievement of desirable results.	Feb, 26

Table 5 Testing Plan